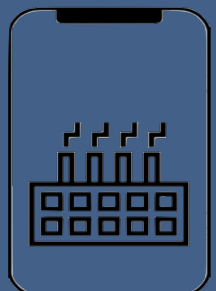
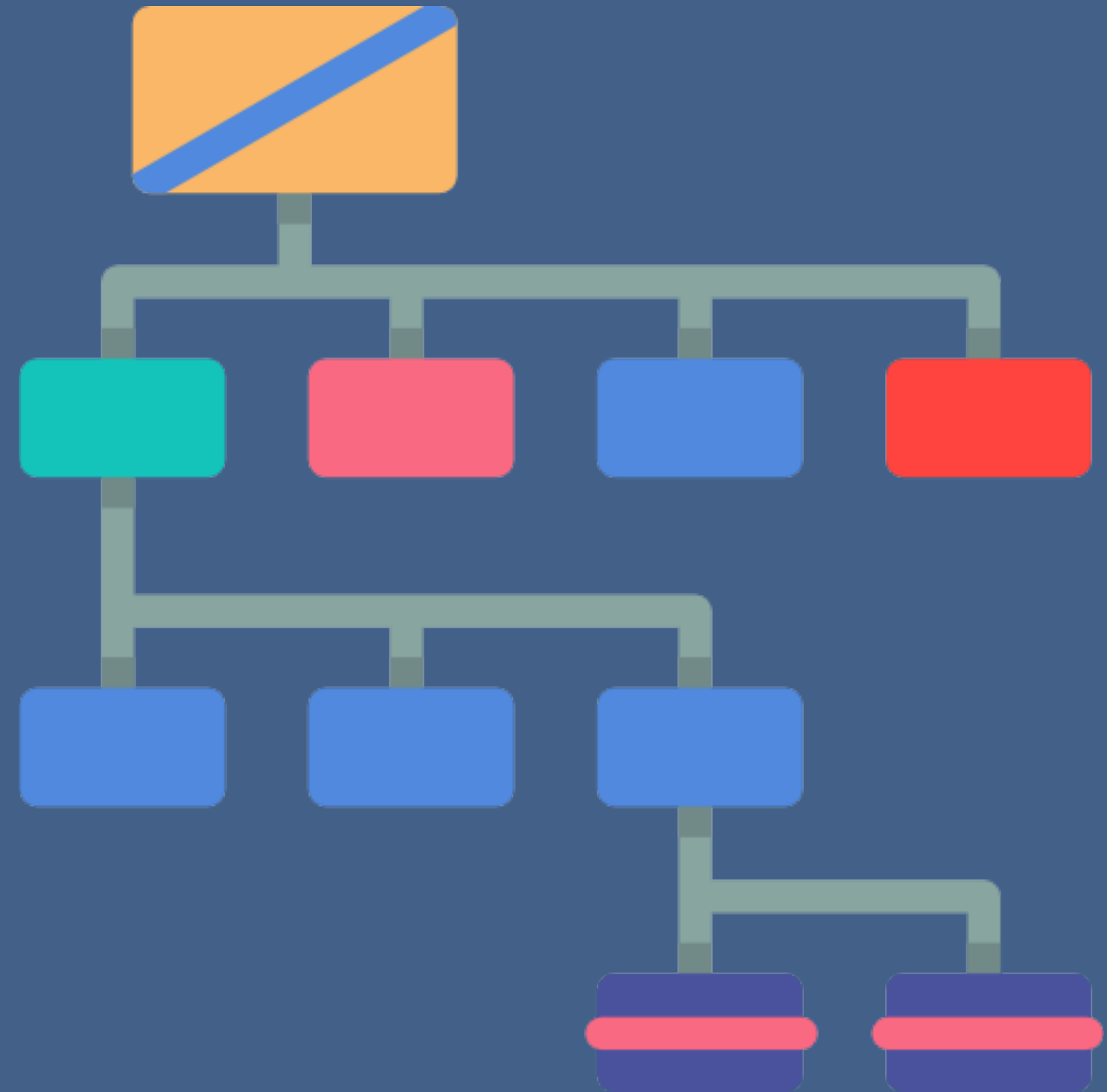
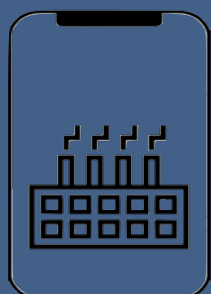
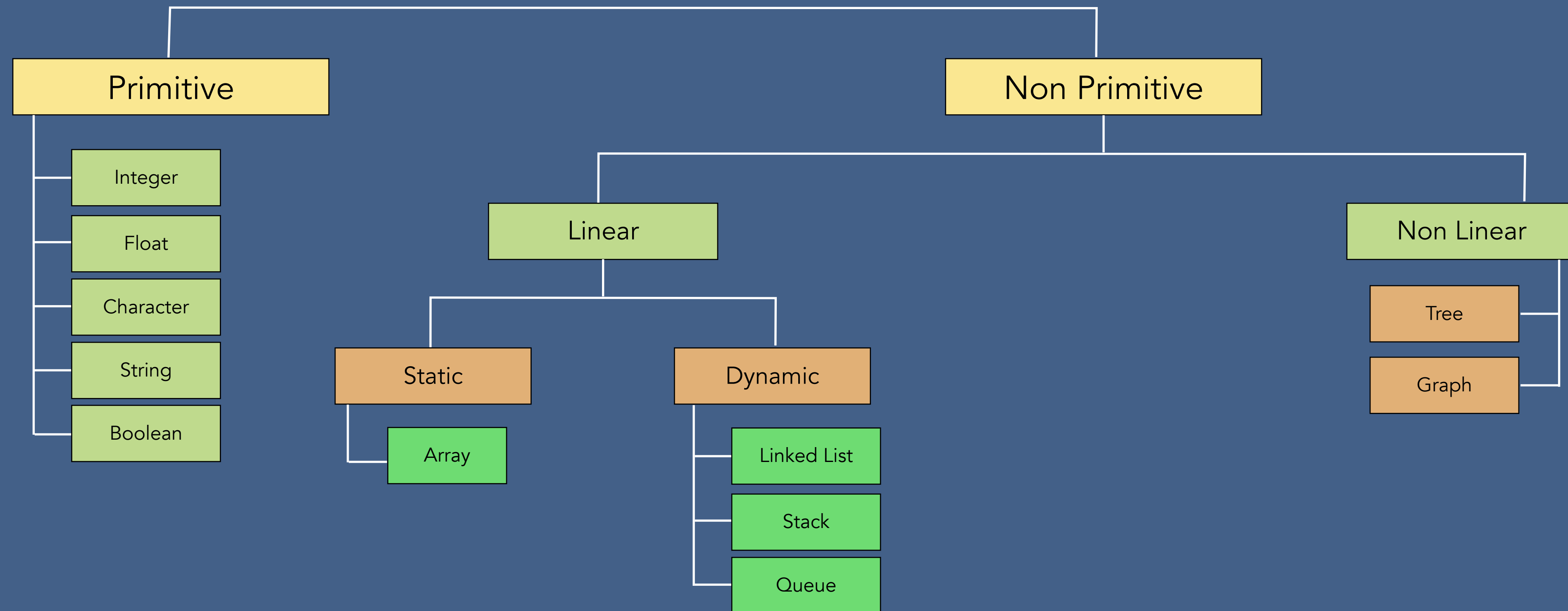


Types of Data Structure

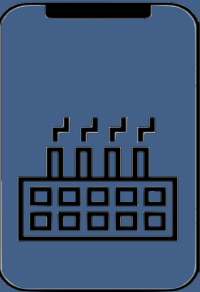


Types of Data Structure

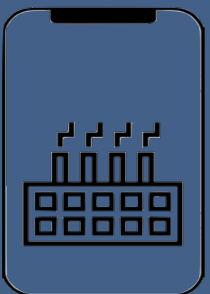
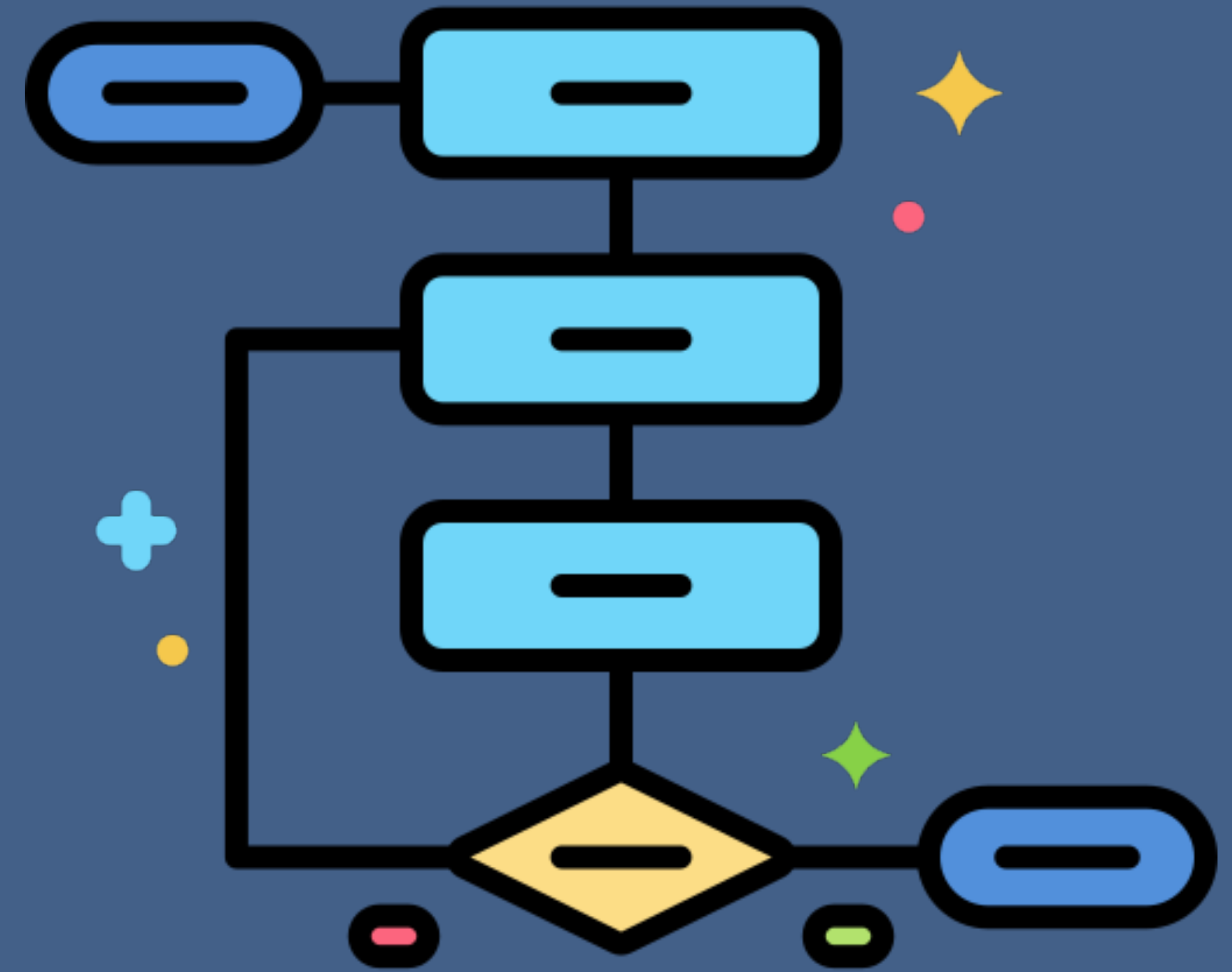


Types of Data Structure

DATA STRUCTURE	Description	Example
INTEGER	Numbers with our decimal point	1, 2, 3, 4, 5,1000
FLOAT	Numbers with decimal point	3.5, 6.7, 6.987, 20.2
CHARACTER	Single Character	A, B, C, F
STRING	Text	Hello, Data Structure
BOOLEAN	Logical values true or false	TRUE, FALSE

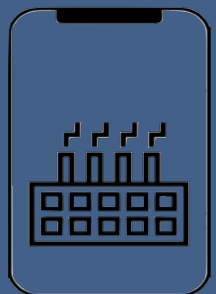


Types of Algorithms



Types of Algorithms

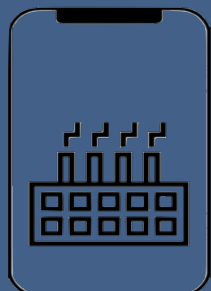
- Simple recursive algorithms
- Divide and conquer algorithms
- Dynamic programming algorithms
- Greedy algorithms
- Brute force algorithms
- Randomized algorithms



Types of Algorithms

Simple recursive algorithms

```
Algorithm Sum(A, n)
    if n=1
        return A[0]
    s = Sum(A, n-1)    /* recurse on all but last */
    s = s + A[n-1]     /* add last element */
return s
```



Types of Algorithms

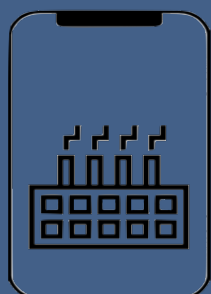
Divide and conquer algorithms

- Divide the problem into smaller subproblems of the same type, and solve these subproblems recursively
- Combine the solutions to the subproblems into a solution to the original problem

Examples: Quick sort and merge sort

Dynamic programming algorithms

- They work based on memoization
- To find the best solution



Types of Algorithms

Greedy algorithms

- We take the best we can without worrying about future consequences.
- We hope that by choosing a local optimum solution at each step, we will end up at a global optimum solution

Brute force algorithms

- It simply tries all possibilities until a satisfactory solution is found

Randomized algorithms

- Use a random number at least once during the computation to make a decision

